

5	(a)	(i)	Resistance is <u>infinite</u> / <u>very high</u>	B1
		(ii)	<u>Resistance decreases as V increases</u> (Ratio of V/I increases)	B1
	(b)	(i)	$R = \rho L/A$ $\rho = (18 \times \pi(0.15 \times 10^{-3})^2/0.94$ $= 1.4 \times 10^{-6} \Omega \text{ m}$	B1

		(ii)	p.d. across wire = $\frac{18}{18 + 2.0}(6.2) = 5.58 \text{ V}$ Since S is at the mid-point, pd across half the length of the wire = $\frac{1}{2}(5.58)$ Voltmeter reading = 2.8 V	C1 A1
		(iii)	Current in the battery: increase Voltmeter reading: decrease	B1 B1
	(c)	(i)	$I = Anvq$ $q = 0.93 / [(\pi(0.15 \times 10^{-3})^2) \times (9.0 \times 10^{28}) \times (1.3 \times 10^{-3})]$ $= 1.1 \times 10^{-19} \text{ C}$	A1
		(ii)	The charge carriers in a metal wire are the electrons which has a charge of $1.6 \times 10^{-19} \text{ C}$. <u>Charge q is less than $1.6 \times 10^{-19} \text{ C}$ the elementary charge.</u> So the value must be wrong	B1

6	(c)	(i)	Since the velocity is always right angle to the magnetic field, by Fleming's left	B1
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